

Session 4 · Deck 4.A1

Root-Finding and Optimization on an Interval

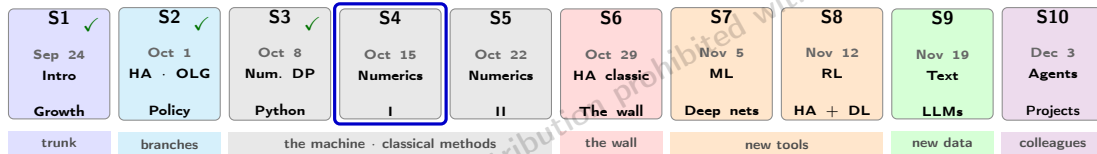
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Thursday, October 15, 2026

Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are

you are here: 4.A1 ▼



Build it on paper → solve it on the machine → push it to the frontier with AI

Block A: 4.A1 root-finding and optimization on an interval · 4.A2 optimization in many dimensions, and where derivatives come from.

Four Methods, One Equation, and No Solution

A market-clearing problem, solved four ways in a notebook:

$$Q^d(p) = 100 - 2p + 10\sqrt{p}, \quad Q^s(p) = -20 + 3p - 0.1p^2.$$

bisection	ValueError
Brent	ValueError
Newton	a number
secant	a number

Excess demand has a **minimum of** +104.7 at $p = 19.3$ and is positive everywhere. There is no equilibrium price. The two open methods return a number anyway, and *which* number depends on the implementation — our run returned negative prices ($p \approx -867$ and -822), where demand is not even defined.

Which two behaved well?

The ones that **refused**. Bisection and Brent need a bracket with a sign change; when you cannot supply one, they tell you so.

That is the whole lesson of the first half of this deck in one slide. A root-finder does not know what a price is. It reports a number; **whether the number means anything is your job**, and the methods differ in how loudly they fail.

Why Economists Spend So Much Time Solving $f(x) = 0$

- **Market clearing:** excess demand = 0 — the price r^* in Aiyagari is exactly this.
- **First-order conditions:** the labour FOC $u_\ell = w u_c$, solved at every state.
- **Euler equations:** 3.A3's time iteration solves one root per grid point per sweep.
- **Any fixed point:** $x = g(x)$ becomes $f(x) = x - g(x) = 0$.

The last line matters: **every fixed-point problem in this course is a root-finding problem**, including the ones we have been solving by iteration.

Two families

Bracketing methods keep an interval known to contain a root. Slow, and they cannot fail.

Open methods follow a local model of f . Fast, and they can run away.

brentq, which 3.A3 used as a sealed box, is a bracketing method wrapped around an open one — which is why it is the default.

Bisection: The One That Cannot Fail

Given $[a, b]$ with $f(a)f(b) < 0$ and f continuous, the intermediate value theorem puts a root inside. Halve, keep the half that still changes sign, repeat.

$$|x_n - x^*| \leq \frac{b - a}{2^n}.$$



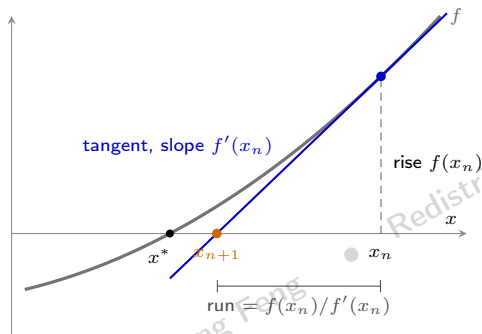
Each row: evaluate f at the midpoint and keep the half whose ends still have opposite signs. The dot is the root; it is never lost.

- **Linear** convergence: one bit per iteration, about 3.3 iterations per decimal digit.
- The bound is *a priori* — you know the iteration count before you start, from the bracket alone.
- It uses nothing about f beyond its sign.

Why it is still the right default

It is the only method here whose guarantee does not depend on where you started, on differentiability, or on luck. Its one requirement is the bracket, and finding one is **your** problem — as the hook showed. Plot the function first.

Newton: Solve the Tangent



Near the root, f looks like its tangent at x_n ,

$$\ell(x) = f(x_n) + f'(x_n)(x - x_n).$$

Newton solves $\ell(x) = 0$ instead of $f(x) = 0$:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)},$$

then repeats from x_{n+1} . Cost per step: one f and one f' .

Read the step as rise over run

Slope = rise $\div$ run, so run = rise $\div$ slope. The rise $f(x_n)$ is how far f is from zero; dividing by the slope turns it into how far to move along the axis. A steep f means a short step, a flat f a long one, and the sign of f/f' points the step towards the zero.

Newton: Why the Error Squares, and Where It Fails

Write $e_n = x_n - x^*$ for the **error**: the signed distance from the current guess to the root.

Expand f around x_n and evaluate it at the root, where $f(x^*) = 0$:

$$0 = f(x_n) - f'(x_n) e_n + \frac{1}{2} f''(\xi) e_n^2$$

for some ξ between x_n and x^* . Divide by $f'(x_n)$ and substitute into $x_{n+1} - x^* = e_n - f(x_n)/f'(x_n)$:

$$e_{n+1} = x_{n+1} - x^* = \frac{f''(\xi)}{2f'(x_n)} e_n^2.$$

The tangent gets the linear term exactly and misses only the curvature term, which is of order e_n^2 . So the new error is a constant times the old error **squared**, and the number of correct digits roughly doubles each step. Measured on the problem of “The Rates, Measured” below:

$$|e_n| = 10^{-1.8} \rightarrow 10^{-4.3} \rightarrow 10^{-9.3} \rightarrow 10^{-15}.$$

Three ways it goes wrong

The argument needs e_n small. Far from the root: $f'(x_n) \approx 0$ makes the tangent nearly flat, so it meets the axis far away · a bad start converges to a different root, or to none · it will happily leave the economically sensible region, as it did on the hook slide.

The standard repair is **damping**: take $x_{n+1} = x_n - \theta f/f'$ with $\theta \in (0, 1]$, backtracking on θ until $|f|$ actually falls.

Secant and Brent: Newton Without the Derivative

Secant. Replace the tangent by the *chord* through the last two points — the slope of f estimated from the history instead of computed:

$$x_{n+1} = x_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}.$$

One new evaluation per step, no derivative, and the order is the golden ratio $\frac{1+\sqrt{5}}{2} \approx 1.618$ — slower than Newton per step, often faster per *second*, because f' is usually the expensive part.

Brent. Keep bisection's bracket as a **safety net**. Inside it, try the fast step (secant, or a parabola through three points); accept it only if it lands inside the bracket and shrinks it enough, and bisect otherwise. Superlinear when the function is nice; only a bounded slowdown against bisection when it is not.

Brent (1973), *Algorithms for Minimization without Derivatives*.

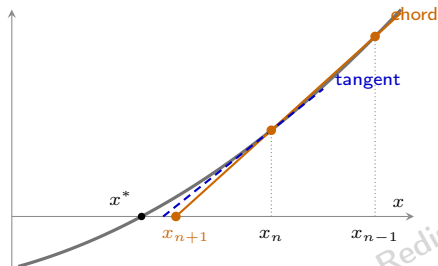
What was in the box

This is brentq. 3.A3 was entitled to trust it because we had *proved* the residual crosses zero exactly once — and a bracketing method with a valid bracket cannot fail.

The general rule: **bracket if you can, and only then go fast**. Every production solver is built this way.

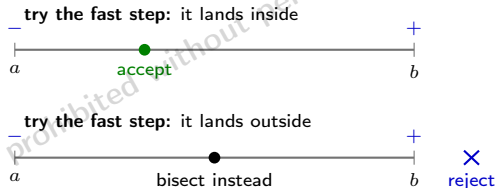
Secant and Brent, in Pictures

Secant: a chord instead of a tangent



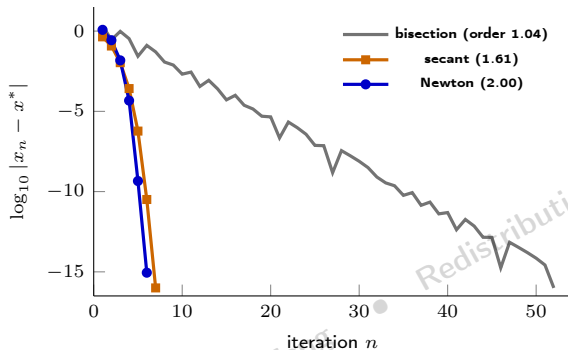
The chord's slope, $\frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}$, is f' estimated from the last two points. Its zero lands close to where the tangent's would, and the closer the two points get, the better the estimate — which is why the order, 1.618, sits just below Newton's 2.

Brent: a fast step inside a safety net



The bracket $[a, b]$ always has a sign change, so the root never leaves it. Each step Brent *tries* the fast point (secant, or a parabola through three points) and keeps it only if it lands inside and shrinks the bracket enough; otherwise it bisection. When f is smooth the fast steps take over; when it is not, you still have bisection's guarantee.

The Rates, Measured



Market clearing for $100p^{-0.5} = e^{0.5p} - 1$, whose root is $p^* = 7.2786$.

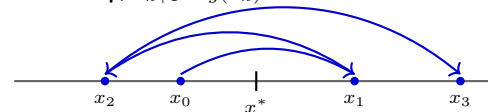
	to 10^{-12}	order
bisection	41	1.04
secant	7	1.61
Newton	6	2.00

The observed orders are 1.04, 1.61 and 2.00 against theoretical 1, 1.618 and 2. The theory is not decoration — it predicts the slopes on the left to two digits.

Observed order estimated from the last clean error triple, $\log(e_{n+1}/e_n)/\log(e_n/e_{n-1})$. Bisection's error is not monotone, so that estimator is noisy for it; its rate is linear with ratio $1/2$ per step, which is the cleaner statement.

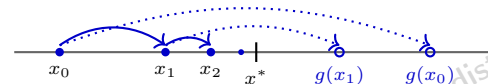
Fixed-Point Iteration Overshoots; Take Only Part of the Step

The full step, $x_{n+1} = g(x_n)$



each landing is further from x^* than the last

Part of the step, $x_{n+1} = (1 - \theta)x_n + \theta g(x_n)$



dotted: where the map points; solid: where you land

Many equilibria are found by iterating $x_{n+1} = g(x_n)$: guess, let the model respond, take the response as the next guess. The map **points the right way**, but a strong response **exaggerates the distance**: the full step lands past x^* , further out than before, and the next swings back further still. **So go only part of the way**:

$$x_{n+1} = (1 - \theta)x_n + \theta g(x_n), \quad \theta \in (0, 1].$$

You land short of where the map points and closer to x^* ; repeat, and the guesses close in. The fixed point itself is untouched: at x^* the map returns x^* , and so does the average.

Too small a θ crawls, so tune it by watching: guesses that *alternate* sides want a smaller θ . The same repair stabilises the outer loops of Krusell–Smith and of transition paths (S6).

The economics: overreaction

Guess K too high $\rightarrow r = F_K$ too low $\rightarrow$ saving too low $\rightarrow$ implied K too low $\rightarrow r$ too high $\rightarrow \dots$ Each swing grows when saving responds strongly; moving K only part of the way settles the loop.

Four Checks Before You Believe a Root

1. **Did it converge?** Read the solver's own flag — do not assume. Both failures on the hook slide reported a number.
2. **Is the residual small?** $|f(x)| < \text{tol}$, checked by you, in the units of the problem.
3. **Is it feasible?** A negative price or a negative consumption is not a solution, whatever the residual says.
4. **Is it unique?** Re-solve from twenty different starts and count the distinct answers.

The failure mode to know

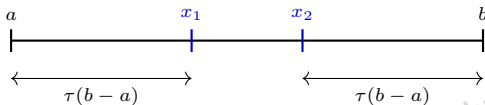
The notebook on the hook slide printed a ✓ next to both wrong answers, because the tick was hard-coded rather than read off `result.converged`. It then *validated* a number that was not a root as the equilibrium.

Check 4 is the one people skip, and it is the one that matters in economics: **multiple equilibria are a result, not a bug** — but only if you looked.

Golden Section: Shrink the Bracket

Problem: find the maximum of f on $[a, b]$ when f has **one peak** there (*unimodal*).

Strategy: narrow the interval by evaluating f at **two** interior points.



Decision rule:

- $f(x_1) > f(x_2)$: the maximum is in $[a, x_2]$ — discard $(x_2, b]$.
- $f(x_1) < f(x_2)$: the maximum is in $[x_1, b]$ — discard $[a, x_1]$.

One peak is what a bracket needs: concave u and concave $\hat{v}$ give the objective $u(y - k') + \beta \hat{v}(k')$ one peak, and linear interpolation keeps $\hat{v}$ concave (4.B1). Fixed costs or discrete choices break it: coarse grid first, then bracket.

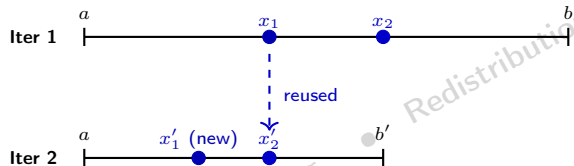
Why two points, not one

Bisection needed one point because the *sign* of f says which side holds a root. One value of f says nothing about which side of the peak you are on. A **comparison** of two values does: with one peak, the lower value cannot sit on the peak's side of the higher one, so that end goes.

The points sit symmetrically, a fraction τ in from each end, so either survivor has the same length; τ is chosen two frames on.

Golden Section: Recycle a Point

The key move: place x_1, x_2 so that after discarding a side, one old point is **already where the next iteration needs a point**.



Here $f(x_1) > f(x_2)$, so $(x_2, b]$ goes; x_2 becomes the new right end, and the old x_1 becomes the new x'_2 . Only x'_1 is new.

The benefit

One new evaluation per iteration, not two. Every step after the first costs a single f , while the bracket keeps shrinking by the same factor.

The question this raises: what τ makes the old x_1 land exactly where the new x'_2 has to be?

Golden Section: Why the Golden Ratio

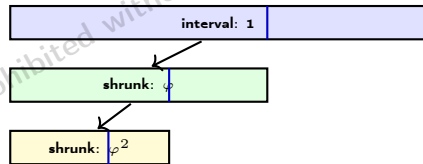
Measure positions as fractions of the interval. The old x_1 sits at τ of $[0, 1]$. After the cut the interval is $[0, 1 - \tau]$, and the new x'_2 must sit at $(1 - \tau)$ of *that*. Recycling requires the two to coincide:

$$\tau = (1 - \tau)(1 - \tau) \implies \tau^2 - 3\tau + 1 = 0,$$

$$\tau = \frac{3 - \sqrt{5}}{2} = 0.382, \quad \varphi = 1 - \tau = \frac{\sqrt{5} - 1}{2} = 0.618.$$

φ is the golden ratio — the *only* number with this property.

Kiefer (1953), "Sequential Minimax Search for a Maximum", *Proc. AMS* 4.



Self-similar: each picture is the previous one scaled by φ . So the interval shrinks by 0.618 per step, one point is always reused, and after n steps the bracket is φ^n of where it began.

The Cost: It Depends on the Tolerance, Not on the Grid

After n evaluations the bracket has length $\varphi^n(b-a)$, so reaching a tolerance ε takes

$$n \approx \frac{\ln(\varepsilon/(b-a))}{\ln \varphi} = O\left(\log \frac{1}{\varepsilon}\right).$$

Measured on $[0, 10]$, maximizing a concave function:

accuracy ε	10^{-4}	10^{-6}	10^{-8}
golden section	26	36	46
theory, $\ln(\varepsilon/10)/\ln \varphi$	23.9	33.5	43.1
grid search	100,000	10 million	1 billion

Each extra digit costs the grid a *factor* of ten and golden section **five more evaluations**.

Brent's minimizer is to golden section what brentq is to bisection: it jumps to the vertex of a parabola through the three best points, and falls back to a golden-section step whenever the parabola misleads.

No M in the formula

The count involves only the tolerance and the width of the interval. It has **nothing to do with the number of grid points M** on which $\hat{v}$ is stored.

The distinction matters: refine your state grid tenfold and the grid search gets ten times more expensive, while golden section costs **exactly the same**.

Takeaways

- Every fixed point in this course is a root-finding problem, and `brentq` is no longer a black box: it is **a bracket with a fast method inside it**. Newton solves the tangent, so its error squares each step: 10^{-12} in six iterations against bisection's 41 — orders 2.00 and 1.04, as the theory says.
- Speed and safety trade off. **Bracket if you can**, then go fast inside the bracket.
- A solver that returns a number has not told you it is right. **Read the flag, check the residual, check feasibility, look for other roots.**
- A fixed-point iteration can overshoot: the map points the right way but exaggerates the distance, so each full step lands further out. **Damping** — moving only part of the way toward $g(x_n)$ — stops the overshoot without moving the fixed point.
- Golden section is bisection for a maximum: **compare two points, discard a side, reuse a point**. The golden ratio is what makes the reuse possible, so each step costs one evaluation.
- Its cost is $O(\log(1/\varepsilon))$ — 46 evaluations to 10^{-8} , where a grid needs a billion points — and **independent of the state grid**. What licenses it is **one peak**, which a concave objective with linear interpolation supplies; with two peaks it can silently return the lower one.

Bridge: When the Choice Is Not a Scalar

Everything here relied on the unknown being **one number on an interval**. Bracketing works because an interval has two ends.

Add labour supply and the household chooses (k', n) . Add a second asset and it chooses (a', b') . Estimate the model and you are choosing a parameter vector. In $\mathbb{R}^N$ there is no bracket to shrink — and the methods that replace it need derivatives, or clever ways of avoiding them.

Next (4.A2)

Optimization in many dimensions: gradients and Hessians, why a Newton step costs $O(N^3)$, Newton and Broyden for systems of equations, BFGS by example, line search, conjugate gradient, Nelder–Mead, simulated annealing, and what changes when constraints bind — then where the derivatives all of this needs come from: finite differences, the step size that is right, and automatic differentiation.